在第一模块，我们多讨论 PDF (probability density function)

该科考试难度 越来越高了。

选择题很简单，大题很难。

而且常常超纲，很搞心态。

于2455 由 Dark Master 制作。

binary detection.

$H \in \{H_0, H_1\}$ 为实际情况.

$\hat{H} \in \{H_0, H_1\}$ 为估计情况.

$\hat{H}\|H$	$H = H_0$	$H = H_1$
$\hat{H} = H_0$	TN	FN
$\hat{H} = H_1$	FP	TP

negative decision $\hat{H} = H_0$ true $\hat{H} = H$

positive decision $\hat{H} = H_1$ false $\hat{H} \neq H$

我们用这个矩阵.

做一个简单的 fish sorting matrix.

$C = 2$ classes $w_1 = \text{salmon}$ $w_2 = \text{sea base}$

$$[n_{ij}] = \begin{bmatrix} n_{11} & n_{12} \\ n_{21} & n_{22} \end{bmatrix} = \begin{bmatrix} 50 & 5 \\ 10 & 35 \end{bmatrix}$$

主对角线 n_{11} n_{22} 为正确判断.

d.h. 50个 salmon 被判断为 salmon.

35个 sea base 被判断为 sea base.

其它为错误判断

5个 sea base 被判断为 salmon.

10个 salmon 被判断为 sea base.

50 + 10 60个 salmon

50 + 5 55个 判断为 salmon.

$\sum_i \sum_j n_{ij} = 100$.

$$[P_{ij}] = \begin{bmatrix} 50\% & 5\% \\ 10\% & 35\% \end{bmatrix}$$

通常我们这样定义 也就是 H与 $\hat{H}$ 的 Joint probability.

$$[P_{ij}] = [P(\hat{w} = w_i, w = w_j)]$$

再进一步. Conditional probability

$$[\bar{P}_{ij}] = [P(\hat{w} = w_i \mid w = w_j)] = \begin{bmatrix} 50/60 & 5/40 \\ 10/60 & 35/40 \end{bmatrix} = \begin{bmatrix} P_{TN} & P_{FN} \\ P_{FP} & P_{TP} \end{bmatrix}$$

$$[P(w = w_j \mid \hat{w} = w_i)] = \begin{bmatrix} 50/55 & 5/55 \\ 10/45 & 35/45 \end{bmatrix}$$

Probability.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

对于任意两个事件 A, B 都成立

Conditional probability.

$$P(A \mid B) = \frac{P(AB)}{P(B)}$$

$P(AB) = P(A \cap B)$ 是 A, B 的 joint probability.

如果 A B 是独立事件. (independent)

$P(A|B) = P(A)$, $P(AB) = P(A) \cdot P(B)$

下面我们需要了解一些 Bayesian theorem.

$$P(w_j \mid \underline{x}) = \frac{P(\underline{x}, w_j)}{P(\underline{x})} = \frac{P(\underline{x} \mid w_j) P(w_j)}{P(\underline{x})} = \frac{P(\underline{x} \mid w_j) P(w_j)}{\sum_{k=1}^{c} P(\underline{x} \mid w_k) P(w_k)}$$

$$\text{posterior} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

$$BR = \sum_{i=0}^{1} \sum_{j=0}^{1} l_{ij} P_{ij}$$

我们在这里做一个 MAP的例子

Classes : $w_1 = \text{Banana}$ $w_2 = \text{Apple}$ $w_3 = \text{Lemon}$

prior : $P(w_1) = 0.3$ $P(w_2) = 0.6$ $P(w_3) = 0.1$

$P(\text{green} \mid w_1) = 0.4$

$P(\text{green} \mid w_2) = 0.3$

$P(\text{green} \mid w_3) = 0.3$

$x = \{\text{green}, \text{yellow}, \text{red}\}$

Which fruit will be sort by color green?

$P(x = \text{green}) = 0.4 \times 0.3 + 0.6 \times 0.3 + 0.3 \times 0.1 = 0.33$.

MAP: $P(w_1 \mid x = \text{green}) = \frac{0.4 \times 0.3}{0.33}$

$P(w_2 \mid x = \text{green}) = \frac{0.3 \times 0.6}{0.33}$ → max. → w_2 : Apple.

$P(w_3 \mid x = \text{green}) = \frac{0.3 \times 0.1}{0.33}$

MAP: max. a posterior.

ML : max. likelihood.

MAP: 已知 prior 和 likelihood.

求得最大的 posterior.

来判断 x 的属性. → $[P_{ij}]$ 的副对角线

此时 ER = $P(\hat{w} \neq w)$ 最小.

$\sum \sum_{i \neq j} P(\hat{w} = w_i \mid w = w_j) P(w_j)$

ML: 仅需要 likelihood.

我们设 $P(w_i) = 1/c$

此时最大的 likelihood 即可做出判断. → $[\bar{P}_{ij}]$ 的副对角线

此时 BER = $\frac{1}{c} \sum \sum_{i \neq j} P(\hat{w} = w_i \mid w = w_j)$ 最小.

下面我们进入第二模块.

Multivariate Gaussian distribution. (多元高斯分布).

$\underline{x} = [x_1, \cdots x_d]^T$ 是一个 random vector.

$$P(\underline{x}) = \frac{1}{(2\pi)^{d/2}\sqrt{|C|}} \cdot e^{-\frac{1}{2}(\underline{x}-\underline{\mu})^T C^{-1}(\underline{x}-\underline{\mu})}$$

$E(\underline{x}) = \underline{\mu}$ 均值向量.

$E[(\underline{x}-\underline{\mu})(\underline{x}-\underline{\mu})^T] = C$ 协方差矩阵.

$\underline{x} \sim N(\underline{\mu}, C)$ $\underline{\mu}$ 是均值, C 是方差.

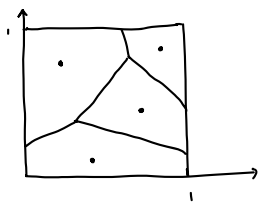
如果不是 C 是 σ^2 $P(\underline{x}) = \frac{1}{(\sqrt{2\pi}\cdot\sigma)^d} \cdot \exp\left(-\frac{1}{2\sigma^2}(\underline{x}-\underline{\mu})^T(\underline{x}-\underline{\mu})\right)$

- Each conditional PDF $p(x_i|x_j)$ is also Gaussian.
- Each affine transform $\underline{y} = A\underline{x} + \underline{b}$ of $\underline{x}$ is again Gaussian with $\underline{y} \sim N(A\underline{y}+\underline{b}, ACA^T)$

MAP decision for Gaussian likelihood - Case A.

- $d=2$. $\underline{x} = [x_1, x_2]^T$ $\underline{x} \mid w_i \sim N(\underline{\mu}_i, C_i)$
- $C_i = \sigma^2 I$
- $P(w_i) = \text{const.}$

$c = 4$ classes



点代表 $\underline{\mu}_i$

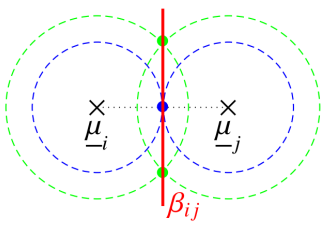
线代表判断边界.

这时, 我们把 MAP 简化为 nearest mean classifier. (最近均值)

这种模型被叫做: Voronoi diagram

$D_{Eucl}(\underline{x}, \underline{\mu}) = \sqrt{(\underline{x}-\underline{\mu})^T(\underline{x}-\underline{\mu})}$

Case A: $\mathbf{C}_i = \mathbf{C}_j = \sigma^2\mathbf{I}$



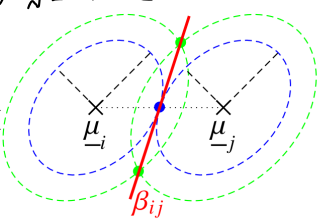
Case - B.

$C_i = C_j \neq \sigma^2 I$.

此时会是两个相同的椭球.

$D_{Maha}(\underline{x}, \underline{\mu}, C) = \sqrt{(\underline{x}-\underline{\mu})^T C (\underline{x}-\underline{\mu})}$

Case B: $\mathbf{C}_i = \mathbf{C}_j \neq \sigma^2\mathbf{I}$

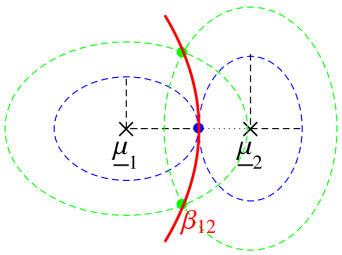


Case - C

$C_i \neq C_j$

两个不相同的椭球.

Case C: $\mathbf{C}_1 \neq \mathbf{C}_2$



Signal model.

- $P_0 = P(H_0)$ $P_1 = P(H_1) = 1 - P_0$ (prior)

likelihood will be like.

$$P(\underline{x}|H_0) = \prod_{n=0}^{d-1} p(x(n)|H_0) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{n=0}^{d-1} x^2(n)\right)$$

$$P(\underline{x}|H_0) = \prod_{n=0}^{d-1} p(x(n)|H_1) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{n=0}^{d-1} (x(n)-A)^2\right)$$

✳ Bayesian decision - 共有 3 种.

① MBR. 需要 $P(\underline{x}|w)$, $P(w)$, L_{ij} → 0/1 loss → MAP (需要 $P(\underline{x}|w)$ 与 $P(w)$) → $P(w)=\frac{1}{C}$ → ML (需要 $P(\underline{x}|w)$)

② minmax 需要 $P(\underline{x}|w)$, L_{ij} ⇝ 用 $P(\underline{x}|w)$ 与 L_{ij} 求出最大 BR 条件下的 $P(w)$, 然后再用 MBR

③ Neyman-Person 需要 $P(\underline{x}|w)$ ⇝ $\max_w P_D$ s.t. $P_{FA} < \alpha$

α 为一个常数 5%, 10%.

DC : A AWGN : σ^2 from d.

$$\frac{\bar{x}}{A} = \frac{1}{Ad}\sum_{n=0}^{d-1} x(n) \gtrless_{H_0}^{H_1} \gamma$$

$$\rho = d \cdot SNR = d\frac{A^2}{\sigma^2} \qquad t(x) = \frac{\bar{x}}{A}$$

* Q-function.

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$x \sim N(0,1)$

CDF : $\phi(x) = \int_{-\infty}^{x} p(z)dz$. $\quad p(x) = \frac{d\phi(x)}{dx}$

$$Q(x) = 1 - \phi(x) = \int_x^{\infty} p(z)dz = 1 - Q(-x)$$

$Q(\gamma)$ 表示 $x > \gamma$ 的概率.

接下来，我想解释一下，用这些来做什么，

我们知道 MAP，需找一个 max，posterior.

(已知 likelihood 与 prior.)

通过 signal model 的方法 H_0 H_1 被我们信号化了. → $P(\underline{x}|H_0)$ $P(\underline{x}|H_1)$

$$P(w=w_i|\underline{x}) = \frac{P(\underline{x}|w=w_i)\,P(w_i)}{P(\underline{x})}$$

max ↑

但此时，A 应是一个向量 $s = [s(0), s(1) \ldots s(d-1)]^T$

也就是说.

$$P(\underline{x}|H_0) = \frac{1}{(2\pi)^{d/2}\sqrt{|C|}} \exp\left(-\frac{1}{2}\underline{x}^T C\,\underline{x}\right)$$

$$P(\underline{x}|H_1) = \frac{1}{(2\pi)^{d/2}\sqrt{|C|}} \exp\left(-\frac{1}{2}(\underline{x}-\underline{s})^T C\,(\underline{x}-\underline{s})\right)$$

$$P_{FA} = \int_{R_1} P(\underline{x}|H_0)\,dx$$

$$P_D = \int_{R_1} P(\underline{x}|H_1)\,dx$$

(NP)

i.i.d. 此时我们简化 MAP decision.

$$\ln\frac{P(\underline{x}|H_1)}{P(\underline{x}|H_0)} \gtrless_{H_0}^{H_1} \ln\frac{P_0}{P_1} \xrightarrow{\text{经过复杂的化简}} \underline{s}^T C^{-1}\underline{s}\left(\frac{\underline{s}^T C^{-1}\underline{x}}{\underline{s}^T C^{-1}\underline{s}} - \frac{1}{2}\right) \gtrless_{H_0}^{H_1} \ln\frac{P_0}{P_1}$$

$$t(\underline{x}) = \frac{\underline{s}^T C^{-1}\underline{x}}{\underline{s}^T C^{-1}\underline{s}} \qquad \rho = \underline{s}^T C^{-1}\underline{s}$$

$$t(\underline{x}) = \frac{1}{\rho}\underline{s}^T C^{-1}\underline{x} \gtrless_{H_0}^{H_1} \frac{1}{\rho}\ln\left(\frac{P_0}{P_1}\right) + \frac{1}{2} = \gamma_{MAP}.$$

$t(\underline{x})|H_0 \sim N(0, 1/\rho)$

$t(\underline{x})|H_1 \sim N(1, 1/\rho)$

$$P_{FA}(\gamma) = P(t \geq \gamma|H_0) = Q(\sqrt{\rho}\gamma)$$

$$P_D(\gamma) = P(t \geq \gamma|H_1) = Q(\sqrt{\rho}(\gamma - 1))$$

假如是 ML decision，

$$P_0 = P_1 = \frac{1}{2}, \quad \gamma_{ML} = \frac{1}{2}, \quad P_{FA} = Q\left(\frac{1}{2}\sqrt{\rho}\right) \quad P_D = 1 - P_{FA}.$$

ratio test

MBR

0/1 loss ↓

$$\frac{P(\underline{x}|H_1)}{P(\underline{x}|H_0)} \gtrless_{H_0}^{H_1} \frac{L_{10}-L_{00}}{L_{01}-L_{11}} \cdot \frac{P(H_0)}{P(H_1)}$$

MAP

equal priors ↓

$$\frac{P(\underline{x}|H_1)}{P(\underline{x}|H_0)} \gtrless_{H_0}^{H_1} \frac{P(H_0)}{P(H_1)}$$

ML

$$\frac{P(\underline{x}|H_1)}{P(\underline{x}|H_0)} \gtrless_{H_0}^{H_1} 1$$

The discriminant functions.

ML decision : $f_i(\underline{x}) = \ln(P(\underline{x}|w_i))$

MAP decision : $f_i(\underline{x}) = \ln(P(\underline{x}|w_i)) + \ln(P(w_i))$

MBR decision : $f_i(\underline{x}) = -R(\hat{w}=w_i|\underline{x}) = -\sum_{j=1}^{c} l_{ij}\cdot P(w_j|\underline{x})$

$$P(w_j|\underline{x}) = \frac{P(\underline{x}|w_j)\cdot P(w_j)}{P(\underline{x})}$$

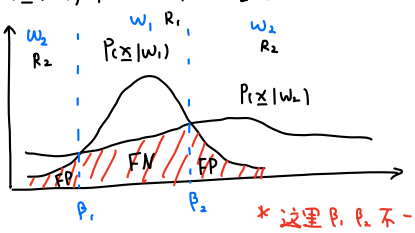
现在只考虑 2 classes 的情况.

$$f(\underline{x}) = f_1(\underline{x}) - f_2(\underline{x})$$

$$f(\underline{x}) \gtrless_{w_2}^{w_1} 0$$

在 $f_1(x) = f_2(x)$ 时取边界 β

然后在 $P(\underline{x}|w_i)$ function 中求 Error rate.



* 这里 β_1 β_2 不一定在交点，只有 ML decision 一定在交点.

$$P_e = \int_{R_2} P(\underline{x}|w_1)P(w_1)\,dx + \int_{R_1} P(\underline{x}|w_2)P(w_2)\,dx$$

= ///

该模型不只在 Gaussian shape 生效.

在任意 shape $\int_{-\infty}^{+\infty} P(\underline{x}|w_i) = 1$ 都生效.

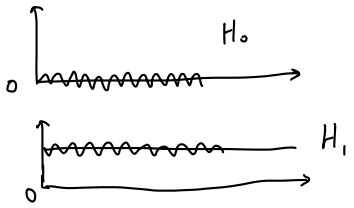
我们进一步理解.

$z(n)$ 表示电吉他播音箱的底噪.

$s(n)$ 表示电吉他输入的声音信号.

$\Rightarrow$ $H_0: x(n) = z(n)$

$H_1: x(n) = z(n) + s(n)$



我们的 Gaussian likelihood $p(\underline{x}|H)$

下面我们初步走向 Next Level.

Machine learning. 这时我们不知道 likelihood $p(\underline{x}|w_i)$

但是我们有 N 个训练样本 (training samples) $\underline{x}_n (1 \le n \le N)$

模型从样本中学习. 再为特征 $\underline{x}$ 分类, 之后拟合边界曲线.

对于训练, 距离 是一个重要概念.

Euclidean distance: $D_{Eucl}(\underline{x}_1, \underline{x}_2) = \sqrt{(\underline{x}_1 - \underline{x}_2)^T (\underline{x}_1 - \underline{x}_2)}$

Mahalanobis distance: $D_{Maha}(\underline{x}_1, \underline{x}_2, C) = \sqrt{(\underline{x}_1 - \underline{x}_2)^T C^{-1} (\underline{x}_1 - \underline{x}_2)}$

这个东西的优势是: ① 他换坐标仍不变.

② 可以更灵活的应对非线性分布.

$\tilde{\underline{x}} = A\underline{x}$

$\tilde{C} = E((\tilde{\underline{x}} - E\tilde{\underline{x}})(\tilde{\underline{x}} - E\tilde{\underline{x}})^T) = ACA^T$

$D_{Maha}(\tilde{\underline{x}}_1, \tilde{\underline{x}}_2, C) = \sqrt{(\tilde{\underline{x}}_1 - \tilde{\underline{x}}_2)^T \tilde{C}^{-1} (\tilde{\underline{x}}_1 - \tilde{\underline{x}}_2)}$

$= \sqrt{(\underline{x}_1 - \underline{x}_2)^T A^T \cdot (ACA^T)^{-1} \cdot A(\underline{x}_1 - \underline{x}_2)} = \sqrt{(\underline{x}_1 - \underline{x}_2)^T C^{-1} (\underline{x}_1 - \underline{x}_2)}$

$= D_{Maha}(\underline{x}_1, \underline{x}_2, C)$

Tipps: $(AB)^T = B^T A^T$

$(AB)^{-1} = B^{-1} A^{-1}$

$-R(w_0) = -l_{00} \cdot P(w_0|\underline{x}) - l_{01} \cdot P(w_1|\underline{x})$

$-R(w_1) = -l_{10} \cdot P(w_0|\underline{x}) - l_{11} \cdot P(w_1|\underline{x})$

w_1 以上 w_0

$l_{10} - l_{00}\ P(w_0|\underline{x}) \gtrless_{w_0}^{w_1} l_{01} - l_{11}\ P(w_1|\underline{x})$

$\frac{P(w_1|\underline{x})}{P(w_0|\underline{x})} \gtrless_{w_0}^{w_1} \frac{l_{10} - l_{00}}{l_{01} - l_{11}}$

$\Downarrow$

$\frac{P(\underline{x}|w_1)}{P(\underline{x}|w_0)} \gtrless_{w_0}^{w_1} \frac{l_{10} - l_{00}}{l_{01} - l_{11}} \cdot \frac{P(w_0)}{P(w_1)}$

下面进入. 训练模型.

Nearest mean

我们以 2 classes 为例.

有 $N = N_1 + N_2$ 个训练对象.

找到 N_1 N_2 的中心 μ_1 μ_2

如果用 D_{Maha} 的话还要求 C_1 C_2

training

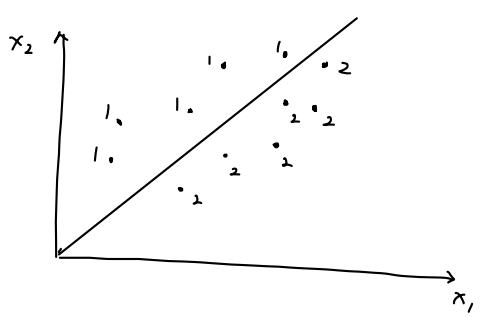
特征 $\underline{x}$

计算 $d_j = D(\underline{x}, \mu_j)$

选择 $y(\underline{x}) = \arg\min_{w_j} d_j$

$(y(\underline{x})$ 是 $\underline{x}$ 的 class $)$

用这些 $\underline{x}$ 去拟合边界线



你的 training samples 至少是 10 times features.

k-Nearest neighbor (kNN)

有 $N = N_1 + N_2$ 个训练对象.

对于 feature $\underline{x}$

计算 $\underline{x}$ 与 N 个训练对象的 $d_n = D_{Eucl}(\underline{x}, \underline{x}_n)$

找到 k 个与 $\underline{x}$ 最近的 $\underline{x}_n$

确定 $y(\underline{x})$ 为这 k 个 $\underline{x}_n$ 中最多的那一类 (class)

→ 拟合边界线.

no training.

Gaussian classifier.

→ plug in 的一种.

plug in 是指一种即定的模型

通过训练来确定其中某些参数.

有 $N = N_1 + N_2$ 个训练对象.

Training:

- likelihood: $N(\underline{\mu}_j, C_j)$
- $\underline{\mu}_j$, C_j
- $\hat{P}(w_j) = \frac{N_j}{N}$

use likelihood ($N(\underline{\mu}_j, C_j)$) and prior ($\hat{P}(w_j)$) in (MAP. ML. MBR. etc.)

→ 拟合边界曲线.

单一高斯几乎只能做线性可分的.

最多做二次可分的

GMM and EM.

GMM 是指多个高斯模型的混合.

EM 是通过迭代计算模型的算法, 可用于估计每个高斯模型的 μ 与 C 和每个高斯模型的比重.

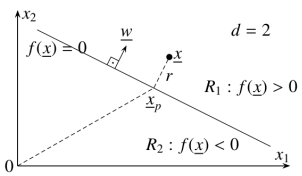
⟹ e.g. class w_1 : $x_1 \sim 0.5\, N(6, 1) + 0.5\, N(12, 1)$

类 1 是两个高斯分布混合构成的.

Hyperplane.

$f(\underline{x}) = f_1(\underline{x}) - f_2(\underline{x}) = \underline{w}^T \underline{x} + \underline{w}_0$

$f(\underline{x}) = 0$ 描述了 Hyperplane. , $(d-1)$.



positive side R_1

negative side R_2.

normal vector $\underline{w}$ points to the positive side.

signed distance.

$r(\underline{x}) = f(\underline{x}) / \|\underline{w}\|$, (distance of $\underline{x}$ to hyperplane · $\mathrm{sgn}(f(\underline{x}))$

Margin. y (1, -1)

$\gamma_n = y_n \frac{f(\underline{x}_n)}{\|\underline{w}\|} \leq 0$, $>/<$, $\underline{x}_n$ correctly / falsely 被 $f(x)$ 定义.

kernel functions. 核函数.

$K(\underline{x},\underline{y}) = \underline{\phi}(\underline{x})^T \underline{\phi}(\underline{y})$ return a large/small value of similar/non-similar feature vectors $\underline{x}, \underline{y}$

相似返回大值，不相似返回小值.

a) Linear kernel.

$K(\underline{x},\underline{y}) = \underline{x}^T \underline{y}$ with $d'=d$.

b) Polynomial kernel. with $d' = \binom{d+m}{m}$:

$K(\underline{x},\underline{y}) = (a + \underline{x}^T\underline{y})^m$, $a \geq 0$, $m \in \mathbb{N}$

c) Radial basis function (RBF) kernel with $d'=\infty$

$K(\underline{x},\underline{y}) = e^{-\gamma\|\underline{x}-\underline{y}\|^2}$, $\gamma > 0$.

γ 越大/小，核函数越 尖锐/平坦，SVM 对特征相似的敏感性越 高/低

tr. 方阵对角线之和.

- $\partial(X_{(\alpha)}Y_{(\alpha)}) = (\partial X_{(\alpha)})\cdot Y_{(\alpha)} + X_{(\alpha)}\cdot(\partial Y_{(\alpha)})$
- $\partial \mathrm{tr}(X_{(\alpha)}) = \mathrm{tr}(\partial X_{(\alpha)})$
- $\partial X_{(\alpha)}^{-1} = -X_{(\alpha)}^{-1}(\partial X_{(\alpha)})\cdot X_{(\alpha)}^{-1}$

- $\frac{\partial}{\partial \underline{x}} \underline{x}^T\underline{a} = \underline{a}^T$
- $\frac{\partial}{\partial \underline{x}} \underline{x}^T B \underline{x} = \underline{x}^T(B+B^T)$
- $\frac{\partial}{\partial \underline{x}} (\underline{x}-\underline{s})^T B (\underline{x}-\underline{s}) = (\underline{x}-\underline{s})^T(B+B^T)$

Hard margin support vector machine.

Given.

$\{(\underline{x}_1, y_1)(\underline{x}_2, y_2)\cdots(\underline{x}_N, y_N)\}$.

Parameter.

$K(\underline{x}_m, \underline{x}_n)$ or γ in case of RBF.

Training.

dual problem

$Q = [q_{mn}]$ with $q_{mn} = y_m y_n K(\underline{x}_m, \underline{x}_n)$

计算 $\underline{\alpha}$.

$$\min_{\underline{\alpha}} \ \frac{1}{2}\underline{\alpha}^T Q \underline{\alpha} - 1^T\underline{\alpha}$$

满足 $\underline{\alpha} \geq \underline{0}$ and $\underline{y}^T\underline{\alpha} = 0$.

Support vectors $SV = \{n \mid \alpha_n > 0\}$

$$w_0 = \frac{1}{|SV|}\left(\sum_{n\in SV} y_n - \sum_{m\in SV}\sum_{n\in SV} a_m y_m K(\underline{x}_m, \underline{x}_n)\right)$$

Classification.

$\underline{x}$.

$$f(\underline{x}) = \sum_{n\in SV} a_n y_n K(\underline{x}_n, \underline{x}) + w_0$$

$$\hat{y}(\underline{x}) = \mathrm{sgn}(f(\underline{x}))$$

$f(\underline{z}) = \underline{w}^T\underline{z} + w_0$

$\underline{w} = \sum_{n\in SV} a_n y_n \underline{z}_n$

$w_0 = \frac{1}{|SV|}\sum_{n\in SV}(y_n - \underline{w}^T\underline{z}_n)$

理解不要间隔.

$$\min_{\underline{w}} \ \frac{1}{2}\|\underline{w}\|^2$$

$$\text{s.t. } y_i(\underline{w}\underline{z}_i + w_0) \geq 1$$

通过这样进行 optimization

是保证对接被超平面正确分开同时还留有至少 "1" 的余地. (鲁棒性)

这样. $\underline{w}\underline{z}_i + w_0 \leq -1 \Rightarrow w_1$

$\underline{w}\underline{z}_i + w_0 \geq 1 \Rightarrow w_2$

这个东西满足 Lagrange function 条件.

$$\min_{\underline{x}} f(\underline{x}) \ \text{s.t. } g_i(\underline{x}) \leq 0 \ (1\leq i\leq p), \ h_i(\underline{x}) = 0 \ (1\leq i\leq q)$$

$$L(\underline{x},\underline{\alpha},\underline{\lambda}) = f(\underline{x}) + \sum_{i=1}^{p} a_i g_i(\underline{x}) + \sum_{i=1}^{q}\lambda_i h(\underline{x}) \Rightarrow$$

$\underline{\alpha}$: Lagrange multipliers $\underline{\lambda}$: dual variables. $\Rightarrow$

KKT conditions:

- stationarity: $\nabla_{\underline{x}} L(\underline{x}^*,\underline{\alpha}^*,\underline{\lambda}^*) = \underline{0}$ $\Rightarrow$
- primal feasibility: $g_i(\underline{x}^*) \leq 0 \ \forall i$ and $h_i(\underline{x}^*) = 0 \ \forall i$
- dual feasibility: $\alpha_i^* \geq 0 \ \forall i$ $\Rightarrow$
- complementary slackness: $\alpha_i^* g_i(\underline{x}^*) = 0 \ \forall i$

1. $\alpha_i^* = 0$: $g_i(\underline{x})$ is inactiv. $g_i(\underline{x}^*) < 0$ $\Rightarrow$
2. $\alpha_i^* > 0$: $g_i(\underline{x})$ is activ $g_i(\underline{x}^*) = 0$.

$$\min_{\underline{w}} \frac{1}{2}\|\underline{w}\|^2 \ \text{s.t. } y_i(\underline{w}\underline{z}_i + w_0) \geq 1.$$

$$\Rightarrow g_i(\underline{x}) = 1 - y_i(\underline{w}\underline{z}_i + w_0) \ (1\leq i\leq p)$$

$$L(\underline{w}, w_0, \underline{\alpha}) = \frac{1}{2}\|\underline{w}\|^2 + \sum_{i=1}^{p} a_i \underbrace{[1 - y_i(\underline{w}\underline{z}_i + w_0)]}_{g_i(\underline{x})}$$

- $\frac{\partial}{\partial \underline{w}} L(\underline{w}, w_0, \underline{\alpha}) = \underline{w} - \sum_{i=1}^{p} a_i y_i \underline{z}_i = 0$

  $\frac{\partial}{\partial w_0} L(\underline{w}, w_0, \underline{\alpha}) = -\sum_{i=1}^{p} a_i y_i = 0$
- $y_i(\underline{w}\underline{z}_i + w_0) \geq 1 \quad (1\leq i\leq p)$
- $a_i \geq 0, \quad (1\leq i\leq p)$
- $a_i[1 - y_i(\underline{w}\underline{z}_i + w_0)] = 0 \quad (1\leq i\leq p)$

SV: $a_i > 0$

SV 是在 红线上的点，只有 SV 能决定解.



$$d(\underline{\alpha},\underline{\lambda}) = \min_{\underline{x}} L(\underline{x},\underline{\alpha},\underline{\lambda})$$

$$d^* = \max_{\underline{\alpha},\underline{\lambda}} d(\underline{\alpha},\underline{\lambda})$$

$$\text{s.t. } \alpha_i \geq 0$$

总结：该类问题需算子 $\underline{\alpha}$，通常的，解 $\underline{\alpha}$ 比较容易.

需解方程数为 n (dual problem)

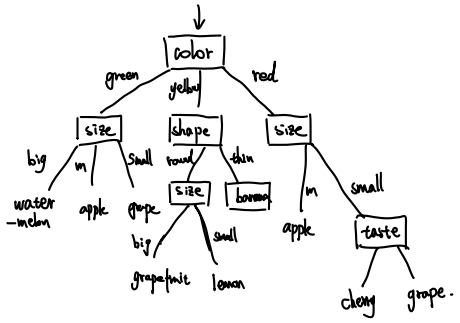
但是，当 d 非常小时，例如 $d=1$

原始问题 (Primal Problem) 变得更容易.

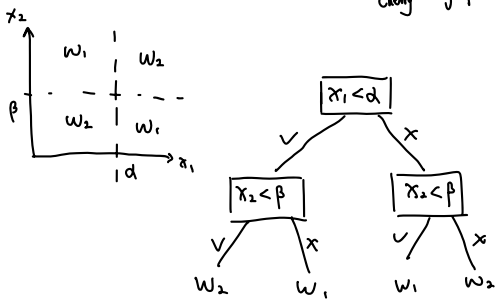
Decision tree. 决策树.

z.B. features { color, size, shape, taste ⋯ }

①



②



RF. random forest.

是一种 对 DT (decision tree) 的集成方法.

采用这种方法可以避免 overfitting.

---

feature dimension reduction 常考概念

How to? 两种方法.

① feature selection

select a subset of $\bar{d}$ features from $\underline{x}$

② feature transform.

find a linear of nonlinear transformation

$\underline{\bar{x}} = \underline{\phi}(\underline{x})$ of $\underline{x}$

Criterion 三种准则.

① filter approach.

Folien 6-2.

② wrapper approach

③ embedded approach

四种模型.

SFFS	feature selection	wrapper approach	supervised
PCA	feature transform	filter approach	unsupervised
LDA	feature transform	filter approach	supervised.
mRMR	feature selection	filter approach	unsupervised.

feature transform

more general, better performance than feature selection.

for the same $\bar{d}$

feature selection = a special case of feature transform.

在之前. 所有的 classifier 都是在已知 label 的情况下进行.

即 supervised learning.

那未知 labels 能不能做呢.

Unsupervised learning. GMM 也可以.

① k-means algorithm.

Given

samples $S = \{\underline{x}_1, \underline{x}_2, \cdots \underline{x}_N\}$

Parameter.

number of cluster C

初始化: 随机选择 C 个中心 $\underline{\mu}_i$ $i \in [1, C]$.

在 S 中. 每个 $\underline{x}$ 最靠近的 $\underline{\mu}_i$, 则判定为 i 类.

然后重新计算类的中心 $\underline{\mu}_i$

循环直到 不再变化结束.

② Fuzzy c-means algorithm.

该算法与上一个类似. 只是在判断 sample 归属时引入了新的算法.

Given

samples $S = \{\underline{x}_1, \underline{x}_2, \cdots \underline{x}_N\}$

Parameter.

number of cluster C, fuzziness $m > 1$

初始化: 随机选择 C 个中心 $\underline{\mu}_i$ $i \in [1, C]$.

$$P_{in} = \frac{\|\underline{x}_n - \underline{\mu}_i\|^{-2/(m-1)}}{\sum_{i=1}^{c} \|\underline{x}_n - \underline{\mu}_i\|^{-2/m-1}}, \quad 1 \le i \le C, \quad 1 \le n \le N.$$

通过这个判断 membership.

重新计算类的中心

$$\underline{\mu}_i = \frac{\sum_{n=1}^{N} P_{in}^m \underline{x}_n}{\sum_{n=1}^{N} P_{in}^m}, \quad 1 \le i \le C$$

重复直到 改变足够小.

---

if C 不知道. mean shift, DBSCAN.

(nonparametric)

차원.

	Supervised	nonlinear
Nearest Mean	✓	×
kNN	✓	✓
NN	✓	✓
SVM	✓	× (feature mapping) ✓

SVM only for 2 classes.

k-Means is not nonlinear.

- D_{Maha} can make quadratic decision bounary.
- D_{Eucl} always only linear

how to do mult. class

① one against one

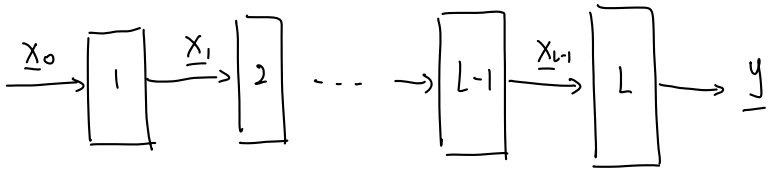
$\frac{C(C-1)}{2}$

w_1 vs w_3
w_2 vs w_3
w_1 vs w_2

② one against others

w_1 vs rest
w_2 vs rest
⋮

Neural Network (NN)



$\underline{x}_L = \phi_L(W_L \underline{x}_{L-1} + b_L)$, $\phi(\cdot)$ is action function always nonlinear.

if linear separable. only need one Layer.
that means no hidden Layer.

$\hat{\underline{y}}_n = f(\underline{x}_n | \underline{\theta})$, loss $L(\hat{\underline{y}}_n, \underline{y}) = \| \hat{\underline{y}}_n - \underline{y} \|^2$

cost function $J = \frac{1}{N} \sum_{n=1}^{N} \| \hat{\underline{y}}_n - \underline{y}_n \|^2$

$\min_{\underline{\theta}} L(\underline{\theta})$.

Soft Margin SVM.

$\min_{\underline{w}} \frac{1}{2} \underline{w}^T \underline{w} + C \sum_i \xi_i$ s.t. $y_i(\underline{w}^T \phi(\underline{x}_i) + w_0) \geq 1 - \xi_i$

C: trade off parameter. $C > 0$
balances wrong decisions against the orginal optimization problem.
SV for $0 < a_n < C$

ξ_i: slack variable.
relax the inequally condition and allow for samples to lie inside the magrin or on the wrong side of decision bounary.
We need them for data set which are not separable. $\xi_i \geq 0$

$\phi(\cdot)$: feature mapping. nonlinear ⟶ linear.
build a nonlinear decision bounary with a linear form.

$K(x, y) = \phi(x) \cdot \phi(y)$